

Case Study: Abnormally High Utilization Rate for Credit Card

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Executive Summary

The analysis revealed an abnormally high number of accounts with extreme utilization rates, in some cases far exceeding industry norms. Python programming was employed to perform the statistical treatments and validate the findings.

Such anomalies are most likely attributable to system or data reporting issues. A system revision and data integrity review should therefore be prioritized. If technical causes are ruled out, the presence of extreme utilization may indicate potential fraud or distressed accounts, in which case escalation to fraud specialists is recommended.

Introduction

Banks rely heavily on interest income as a core revenue stream. In the context of credit cards, a high utilization rate typically signals strong potential for interest earnings. While occasional instances of unusually high utilization are expected and generally benign, the latest analysis reveals that 8.8% of the customer base is exhibiting abnormally high utilization rates.

This level of deviation is far beyond industry norms and raises significant concerns. Such anomalies may stem from system or data reporting issues, but they could also indicate potential fraud or distressed accounts. Immediate investigation and resolution are therefore necessary to safeguard both portfolio health and customer trust.

Background of the Study

Credit card utilization rates typically range from 0% up to 100% of the assigned credit limit. Higher utilization rates are generally favorable for banks in terms of revenue generation, as they increase the potential for interest income. However, elevated utilization also signals greater credit risk.

Industry practice recognizes that the optimal utilization range for balancing profitability and risk is approximately 30% to 70%. Values below this range tend to generate limited revenue, while values above it increases the likelihood of delinquency.

In rare cases, outstanding balances may exceed the credit limit. Such exceptions are usually the result of issuer-granted approvals or accrued charges, and they typically fall within 101% to 120% of the credit limit. These scenarios are considered manageable but warrant close monitoring to ensure they do not escalate into systemic risk.

Beyond this threshold, abnormal utilization rates (greater than 120%) can distort portfolio risk metrics, misrepresent customer behavior, and potentially mask fraudulent or distressed accounts. Such anomalies not only undermine the accuracy of credit risk models but also pose reputational and financial risks if left unresolved.

Methodology

First, necessary python libraries were imported and loaded the dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats

pd.set_option('display.max_columns', 100)
sns.set_style('whitegrid')

df = pd.read_csv('mockup_credit_card_excel_1000_rows.csv')
```

Quick data validation using the info() method provided the following overview:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   customer_id           1000 non-null   int64
 1   age                   1000 non-null   int64
 2   segment               1000 non-null   object
 3   region                1000 non-null   object
 4   tenure_months         1000 non-null   int64
 5   credit_limit           1000 non-null   int64
 6   avg_monthly_spend     1000 non-null   float64
 7   txns_last_3mo         1000 non-null   int64
 8   days_past_due         1000 non-null   int64
 9   churn_flag            1000 non-null   int64
10   current_balance       1000 non-null   float64
11   utilization_rate       1000 non-null   float64
12   pct_revolve           1000 non-null   float64
13   revolve_balance       1000 non-null   float64
14   delinquent_flag       1000 non-null   int64
dtypes: float64(5), int64(8), object(2)
memory usage: 117.3+ KB
```

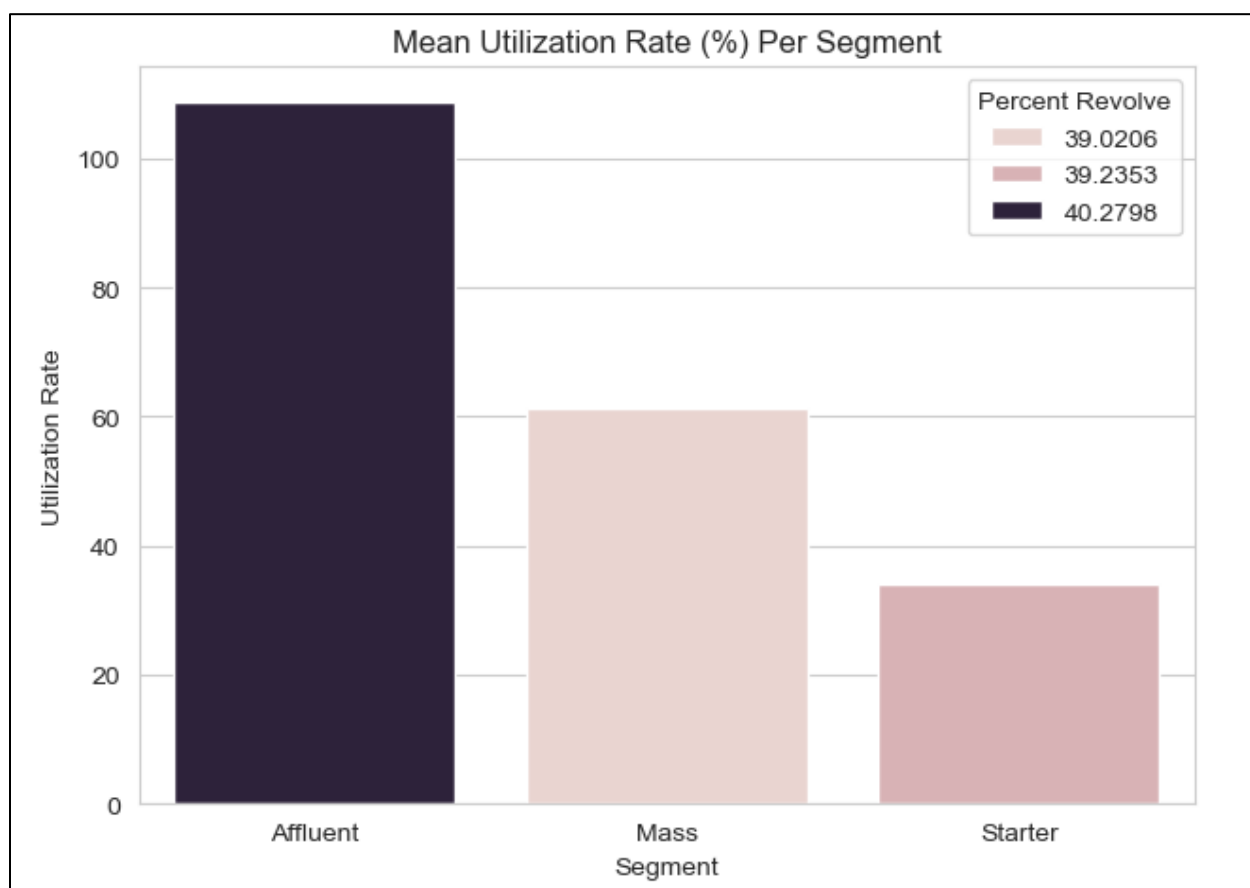
All columns are correctly typed, with zero missing values. The dataset is tidy and therefore ready for analysis.

Exploratory Data Analysis

Quick descriptive statistics using *describe()* method gives us:

	customer_id	age	tenure_months	credit_limit	avg_monthly_spend	txns_last_3mo	days_past_due	churn_flag	current_balance	utilization_rate	pct_revolve	revolve_balance	delinquent_flag
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	100500.500000	45.379000	91.795000	67520.000000	17948.161000	21.990000	4.350000	0.12100	27709.239000	0.630640	0.393220	10979.222000	0.089000
std	288.819436	14.122138	52.230403	43395.077287	8564.553391	4.701277	15.362731	0.32629	17931.719874	0.676334	0.193573	9602.999673	0.284886
min	100001.000000	21.000000	1.000000	20000.000000	1000.000000	10.000000	0.000000	0.00000	0.000000	0.000000	0.020000	0.000000	0.000000
25%	100250.750000	33.000000	47.000000	20000.000000	11746.750000	19.000000	0.000000	0.00000	14478.750000	0.210000	0.240000	3791.250000	0.000000
50%	100500.500000	46.000000	93.000000	50000.000000	18033.500000	22.000000	0.000000	0.00000	25449.000000	0.410000	0.370000	8708.000000	0.000000
75%	100750.250000	57.000000	139.000000	80000.000000	23723.750000	25.000000	0.000000	0.00000	38043.250000	0.830000	0.540000	14830.250000	0.000000
max	101000.000000	69.000000	179.000000	200000.000000	45566.000000	38.000000	90.000000	1.00000	99207.000000	4.960000	0.950000	61180.000000	1.000000

Upon review, the maximum value observed in the *utilization_rate* column is 4.96. This figure is highly unusual and falls well outside expected ranges. Such an anomaly is most likely the result of an encoding or system error, rather than genuine customer behavior.

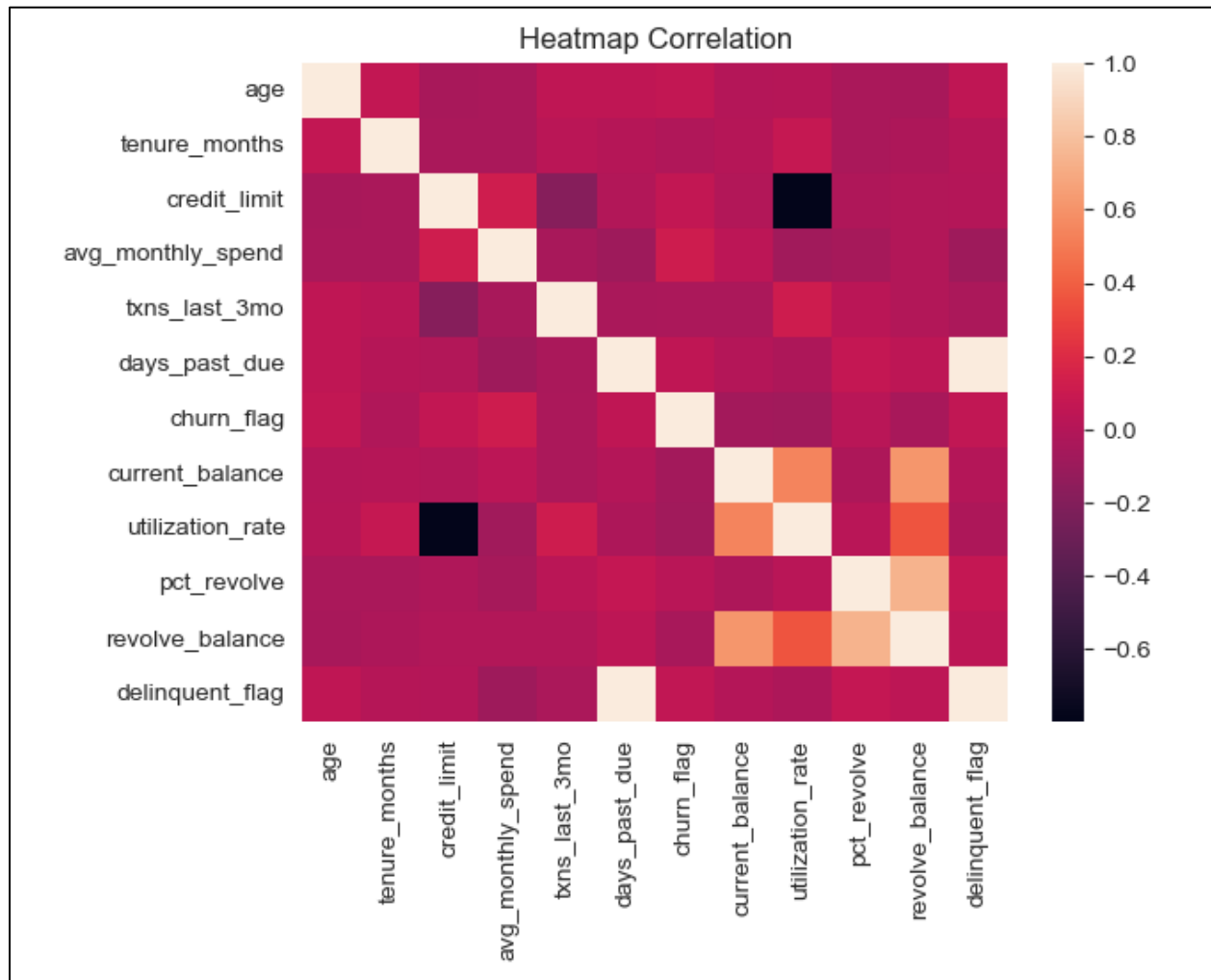


As observed, the Affluent segment records the highest mean utilization rate and percent revolve. This outcome is highly unlikely in practice and, if validated, represents a potential red flag. Typically, affluent credit users maintain utilization around 30% of their credit limit to demonstrate responsible borrowing behavior. While individual customers may occasionally exceed this benchmark, it is far more improbable for the segment's mean values to reflect such extremes.

Assuming the dataset has been fully validated, this anomaly raises an important question: What underlying factors are driving this behavior?

Findings and Analysis

To identify factors influencing the credit utilization rate of Affluent customers, the Spearman Correlation Coefficient was calculated between utilization_rate and other variables. Spearman was chosen as the method since its assumptions were satisfied. Alongside correlation coefficients, p-values were also computed to validate that the observed relationships are statistically significant rather than coincidental.



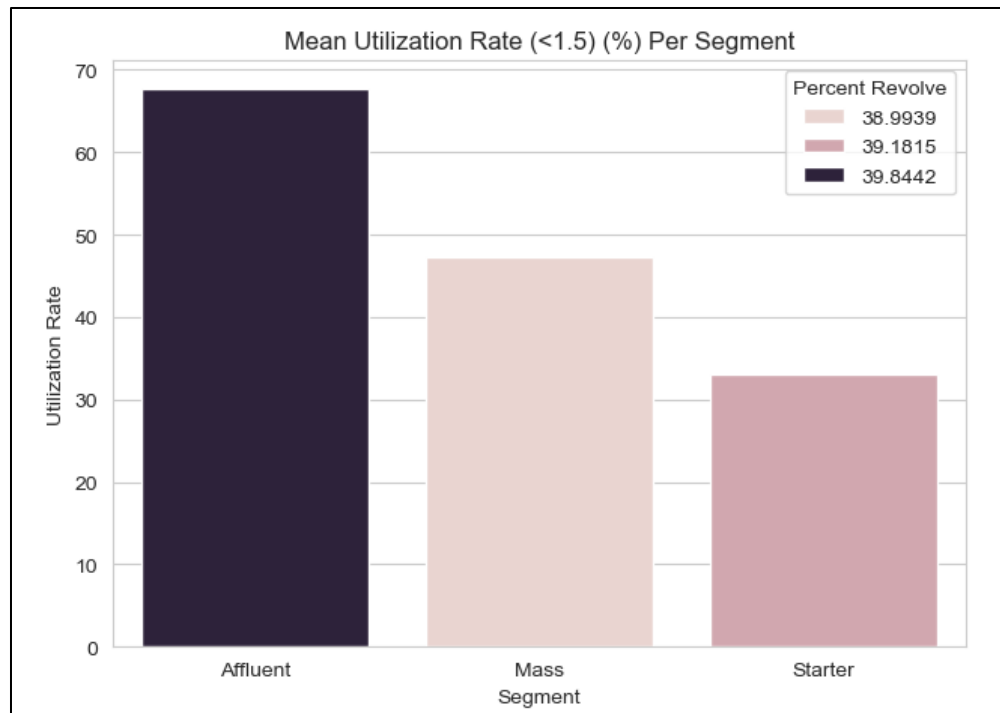
	age	tenure_months	credit_limit	avg_monthly_spend	txns_last_3mo	days_past_due	churn_flag	current_balance	utilization_rate	pct_revolve	revolve_balance	delinquent_flag
age	0.0	0.7275	0.7298	0.0346	0.5302	0.9816	0.6224	0.0698	0.0614	0.7281	0.0625	0.9508
tenure_months	-0.011	0.0	0.7484	0.8659	0.7343	0.649	0.3511	0.6183	0.7338	0.04	0.2421	0.6257
credit_limit	-0.0109	0.0102	0.0	0.3076	0.0542	0.3166	0.7897	0.0305	0.0	0.3699	0.0779	0.2779
avg_monthly_spend	-0.0668	-0.0053	0.0323	0.0	0.0802	0.5179	0.0775	0.8987	0.5142	0.3475	0.9215	0.5116
txns_last_3mo	-0.0199	0.0107	-0.0609	-0.0553	0.0	0.0686	0.8027	0.8232	0.4478	0.0953	0.5439	0.0668
days_past_due	-0.0007	0.0144	0.0317	-0.0205	0.0576	0.0	0.556	0.2607	0.2421	0.2187	0.8473	0.0
churn_flag	0.0156	-0.0295	0.0084	0.0558	-0.0079	-0.0186	0.0	0.2495	0.2374	0.9492	0.5212	0.5474
current_balance	-0.0574	-0.0158	0.0684	0.004	-0.0071	-0.0356	-0.0364	0.0	0.0	0.4677	0.0	0.2615
utilization_rate	-0.0592	-0.0108	-0.6133	-0.0207	0.024	-0.037	-0.0374	0.6876	0.0	0.7473	0.0	0.2251
pct_revolve	-0.011	-0.0649	0.0284	-0.0297	0.0528	0.0389	0.002	0.023	0.0102	0.0	0.0	0.2293
revolve_balance	-0.0589	-0.037	0.0558	0.0031	0.0192	0.0061	-0.0203	0.7724	0.5535	0.5858	0.0	0.8519
delinquent_flag	0.002	0.0154	0.0343	-0.0208	0.058	0.9988	-0.019	-0.0355	-0.0384	0.038	0.0059	0.0

Note: In the correlation matrix, the upper triangle (including the diagonal) displays the p-values, while the lower triangle contains the Spearman correlation coefficients. This dual representation allows both the strength and significance of relationships to be assessed briefly.

Observations:

- **Utilization Rate and Current Balance**
 - These variables are moderately correlated, which is expected since utilization rate is defined as current balance ÷ credit limit. Naturally, as current balance increases, utilization rate also rises.
- **Revolve Balance and Utilization Rate**
 - A moderate correlation is also observed here. Revolve balance is calculated as current balance × percent revolve, and since current balance itself is proportional to utilization rate, increases in utilization rate led to corresponding increases in revolve balance.
- **Credit Limit and Utilization Rate**
 - A strong inverse correlation exists between credit limit and utilization rate. This is inherent in the formula, as utilization rate decreases when credit limits are higher, holding balances constant.
- **Other Variables**
 - The remaining correlations across variables are weak or negligible, indicating limited direct relationships.

Analysis conducted after removing records where *utilization_rate* ≥ 1.5:



Observations:

- The revised graph appears more reliable, with the mean utilization rate falling within a reasonable range.
 - Both the mean utilization rate and percent revolve are positioned within the sweet spot for profitability:
 - Utilization Rate - Below 30% implies low profitability (customers pay in full, little interest income). Above 70% implies high delinquency risk and regulatory concern.
 - Percent Revolve - Too low (<30%) implies most customers are transactors, generating little interest. Too high (>70%) → signals financial stress, higher default risk.
 - Together, these ranges represent the optimal profitability zone for banks.

Analysis performed on records where *utilization_rate* ≥ 1.5:

A subset of the data where *utilization_rate* ≥ 1.5 was extracted for closer examination.

	customer_id	age	segment	region	tenure_months	credit_limit	avg_monthly_spend	txns_last_3mo	days_past_due	churn_flag	current_balance	utilization_rate	pct_revolve	revolve_balance	delinquent_flag
9	100010	31	Affluent	South	21	20000	20734.0	19	0	0	31783.0	1.59	0.40	12713.0	0
11	100012	44	Mass	South	117	20000	18156.0	17	0	0	43722.0	2.19	0.20	8744.0	0
15	100016	23	Affluent	West	39	20000	32471.0	22	0	0	34818.0	1.74	0.25	8704.0	0
16	100017	42	Mass	South	137	20000	19720.0	20	0	0	66790.0	3.34	0.47	31391.0	0
26	100027	42	Mass	North	38	20000	1280.0	19	0	0	31281.0	1.56	0.56	17517.0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
860	100861	33	Affluent	South	86	20000	17257.0	16	0	0	58307.0	2.92	0.36	20991.0	0
877	100878	52	Mass	North	70	20000	24467.0	18	0	0	31884.0	1.59	0.39	12435.0	0
892	100893	25	Affluent	East	102	20000	15620.0	22	0	0	89287.0	4.46	0.55	49108.0	0
926	100927	44	Mass	East	74	20000	44792.0	19	0	0	42689.0	2.13	0.45	19210.0	0
930	100931	34	Mass	North	170	20000	22702.0	23	0	0	55718.0	2.79	0.30	16715.0	0

88 rows × 15 columns

Observations:

- It is highly unlikely that the anomalies are due to human or encoding error.
 - Research suggests average manual data entry error rates are ~1%.
 - With 88 rows affected (8.8% of the dataset), the scale of anomalies far exceeds typical human error margins.
- If this is not an error, then it should be investigated immediately as there is a very high chance that this is a fraud case.

Recommendations

- Conduct an immediate **internal investigation** into the data collection process to rule out system or human error.
- If errors are excluded, proceed with a **revision of services and workflows**.
 - There may be loopholes in current processes that allow customers to exceed credit limits, exposing the bank to heightened risk.
- Allowing customers to exceed limits undermines trust with investors, regulators, and customers.
- Over-limit balances can trap customers in debt cycles, leading to complaints, attrition, and reputational risk.